The Sharp Benchmark Constant for the Deng–Hani–Ma Two-Layer Cutting Algorithm

Zhangjinba

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Abstract

We consider the two-layer Toy I molecule model arising in the combinatorial part of the Deng–Hani–Ma hard-sphere argument. For every finite Toy I molecule and every permitted execution of their six-step cutting algorithm, we prove that the number of good two-atom components is at least $\lceil (|M_U| - 1)/3 \rceil$, improving the source benchmark $(|M_U| - 1)/5$. The proof is an amortised flow argument: orphaned cross-bonds are paid for by persistent fixed-end tokens created by earlier cuts. Three explicit path families attain the bound for every $|M_U| \geqslant 4$, so the benchmark constant is exact. As a scoped consequence, the intrinsic score satisfies $v_d(M) \geqslant \lceil \rho(M)/3 \rceil - 10d$ within Toy I. No claim is made for larger toy classes, arbitrary legal cuts, or the global molecule problem.

Mathematics Subject Classifications: 05C05, 05C85, 05C90

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1 Source contract and scope

We use the two-layer model and Toy I in Section 11 of the public arXiv version of Deng–Hani–Ma [1, Definitions 11.1–11.4]. Each atom has four slots, partitioned into bonds, free ends, and fixed ends; fixed ends do not count toward degree. There are at most two parent slots and at most two child slots. A cross-layer bond is directed from M_U to M_D and occupies a parent slot of its lower endpoint. Cutting a connected set as free turns each boundary bond into one persistent fixed end at the surviving endpoint (bottom at a surviving parent and top at a surviving child); fixed ends are neither rewired nor duplicated. Both layers are trees with no initial fixed ends; no atom of M_D is a parent of an atom of M_U; and every atom of M_U has exactly one bond to M_D.

Independent Researcher.

The residual M_D is proper in the source sense. We run the six-step algorithm of Definition 11.4, allowing all its tie-breaks. Proposition 11.7 of the same source gives a complete legal run, elementary output, and $\#\{4\} = 1$.

Call a complete execution of the six-step algorithm, with any permitted tie-break and an empty final residual molecule, a *Definition-11.4 execution*. Every cut in such an execution is an elementary component; the prescribed three-atom $\{343\}$ cleanup macro is shorthand for its split into one $\{3\}$ and one $\{33\}$, so it contributes one to $\#\{33\}$ and none to $\#\{4\}$. Separately, a *legal complete cut* means any sequence of elementary cuts that ends with an empty residual molecule. The intrinsic value v_d maximises the resulting score over these general legal complete cuts, whereas the benchmark theorem below concerns only Definition-11.4 executions.

For a benchmark run, let $T = \#\{33\}$ and $n = |M_U|$, and define

$$a_n = \min_{|M_U|=n} \ \min_{\text{Definition-11.4 tie-breaks}} T,$$

where the first minimum ranges over finite Toy I molecules. The source gives $\#\{4\} = 1$ and $T \geqslant (n-1)/5$. Our result replaces only the second estimate.

The intrinsic quantities are

$$v_d(M) = \max_C \bigl(\#\{33\}(C) - 10d\,\#\{4\}(C)\bigr), \qquad W(M) = \max_C \#\{33\}(C),$$

where C ranges over legal complete cuts. The benchmark theorem addresses a_n; the scoped transfer to v_d is in Section 4.

1.1 Source and related work

Deng–Hani–Ma supply the original 1/5 benchmark estimate and its role in the hard-sphere derivation [1]. The survey of Bodineau–Gallagher–Saint-Raymond–Simonella reproduces the surrounding reduction [2]. Bruned–Clarisse study a Kruskal-style reduction for a wave-molecule problem [3]; that is a structural result on the wave side, not the quantitative hard-sphere count proved here. Our claim is therefore a quantitative Toy I benchmark result, not an algorithm-classification claim or a new kinetic-limit theorem. The literature search was bounded, so no exhaustive priority claim is made.

2 Exact accounting

Call operation (a) an A event, operation (b) a B event, a cleanup of an adjacent degree-three pair a C event, a cleanup of a $\{343\}$ triple a D event, and a one-atom upper-layer removal after its cross-bond partner has gone a U event. Let $E_<$ be the B, C, D events whose pre-cut state still contains at least one upper-layer atom. Cleanups after the upper layer is empty are excluded: they cannot orphan a future upper-layer partner.

For a complete run, the counting identities are

$$T = B + C + D, \qquad A - (C + D) = 1, \qquad A + B + U = n. \tag{2.0}$$

The first counts the good components produced by (b), C, and the D split; the second is the change of the source potential; the third partitions the upper-layer atoms.

For $e \in E_<$, let $S_e \subset M_D$ be its lower-layer cut set. Define q_e as the live cross-bonds removed by e, except that for a B event the selected upper–lower bond is omitted. Define s_e as the number of live lower-layer boundary arcs $p \to v$ with $p \notin S_e$ and $v \in S_e$, counted by edges, and let r_e be the number of degree-three atoms in S_e carrying a bottom fixed end.

Thus q_e counts future upper atoms orphaned by the event, while each s_e creates one bottom-fixed-end token at the surviving lower parent.

Lemma 1 (Exact orphan partition)**.** *For every complete run,*

$$\#\mathrm{CUT_U} = \sum_{e \in E_<} q_e. \tag{2.1}$$

Proof. Every $\mathrm{CUT_U}$ atom has lost its unique initial cross-bond partner earlier. A C or D event removes exactly the live cross-bonds counted by q_e. A B event removes its selected upper atom together with the selected bond; its other live cross-bonds are exactly those counted by q_e. Conversely, every counted bond has a still-present upper endpoint and produces one later $\mathrm{CUT_U}$. Events after the upper layer is empty have no live cross-bond to a future upper atom. □

Lemma 2 (Local flow capacity)**.** *For every $e \in E_<$,*

$$q_e + s_e \leqslant 2 + r_e. \tag{2.2}$$

Proof. For a degree-three lower atom v, let c_v be its live cross-bonds, p_v its live lower-layer parents, and let $t_v \in \{0, 1\}$ indicate whether its unique fixed end is bottom. The atom has two parent slots; a top fixed end occupies one of them and a bottom fixed end does not. Hence

$$c_v + p_v + (1 - t_v) \leqslant 2. \tag{2.3}$$

For B, $q_e = c_v - 1$, $s_e = p_v$, and $r_e = t_v$, giving the stronger $q_e + s_e \leqslant r_e$. For C, one internal lower bond contributes one parent slot, so summing (2.3) gives $q_e + s_e \leqslant 1 + r_e$. For D, the two outer atoms have degree three, the centre has degree four and no fixed end, and the two internal bonds contribute two parent slots. Summing gives $q_e + s_e \leqslant 2 + r_e$. □

Lemma 3 (Bottom-token injection)**.**

$$\sum_{e \in E_<} r_e \leqslant \sum_{e \in E_<} s_e. \tag{2.4}$$

Proof. Initially the lower layer has no fixed ends. Before an event in $E_<$, the only operations removing lower atoms are earlier B, C, D events. Operations (a) and U remove only upper atoms and create top, not bottom, fixed ends on the lower side; step (6) occurs only after the upper layer is empty. Thus each bottom fixed end counted by an r_e has a unique earlier boundary arc counted by some s_f. The fixed end is never rewired or duplicated, so distinct consumed tokens have distinct source arcs. □

Theorem 4 (Sharp Toy I benchmark bound)**.** *For every finite Toy I molecule and every legal tie-break of Definition 11.4,*

$$\#\{33\} \geqslant \left\lceil \frac{|M_U|-1}{3} \right\rceil. \tag{2.5}$$

Proof. Summing (2.2) and using (2.4) gives

$$\sum q_e \leqslant 2|E_<| \leqslant 2\#\{33\}.$$

The first inequality uses cancellation of the s_e terms, and the second uses that every B, C, D event in $E_<$ produces one good component. By (2.1), $\#\mathrm{CUT_U} \leqslant 2\#\{33\}$. From (2.0), eliminating the event counts gives

$$\#\{33\} = |M_U| - 1 - \#\mathrm{CUT_U}.$$

Therefore $|M_U| - 1 \leqslant 3\#\{33\}$, and integrality gives (2.5). □

The exact run-level contract is

$$\sum_{\text{cleanup } k} b_k + \sum_{B \text{ events } e} q_e \leqslant 2(B + C + D).$$

The older condition $\sum b_k \leqslant 2(C + D)$ is a stronger sufficient condition; it is neither used nor claimed here.

3 Sharpness: matching explicit families

The three families below are finite Toy I molecules. Their lower layers are paths and their upper layers are paths; the cross-bond maps are given explicitly so that the construction is independent of enumeration.

Transpose family. For $k \geqslant 1$, take $n = 3k + 1$, let $M_U = (u_0, \ldots, u_{3k})$ and $M_D = (d_0, \ldots, d_{3k-1})$ be directed paths with internal bonds $u_i \to u_{i+1}$ and $d_j \to d_{j+1}$, and for $0 \leqslant i < 3k$ write $i = qk + r$, $q \in \{0, 1, 2\}$, $0 \leqslant r < k$, and join u_i to d_{3r+q}. Add $u_{3k} \to d_0$. The lowest upper atoms are removed in the order $u_{3k}, u_{3k-1}, \ldots, u_{2k}$, giving $k+1$ operations (a). Their lower partners are d_0, d_{3r+2}. After these cuts the degree-three set is $\{d_0\} \cup \{d_{3r+2} : 0 \leqslant r < k\}$; the points are separated along the path until the final cut, when d_0, d_2 and their degree-four neighbour d_1 form the first $\{343\}$. The blocks $\{d_{3r}, d_{3r+1}, d_{3r+2}\}$ can then be removed successively in the indicated legal tie-break. For $0 \leqslant r < k - 1$, after block r is removed, d_{3r+3} becomes degree three, d_{3r+5} was already degree three, and d_{3r+4} remains degree four because its cross-bond comes from the still present u_{k+r+1}. In block r, the three designated cross-bonds come from u_r, u_{k+r}, u_{2k+r}. For $r \geqslant 1$, only the last of these endpoints has already been removed, so the cleanup breaks two live cross-bonds. For $r = 0$, there is in addition $u_{3k} \to d_0$, but both this endpoint and the designated endpoint u_{2k} have already been removed; again exactly two live cross-bonds are broken. The final block $r = k - 1$ is the terminal cleanup, so no successor indices are needed. This proves inductively that the block cascade is the stated one. After the k cleanups, the remaining $2k$ upper atoms are removed by $\mathrm{CUT_U}$. Thus

$$(A, B, C, D, U) = (k + 1, 0, 0, k, 2k), \qquad \#\{33\} = k.$$

Padded family. Starting from the transpose family, append one atom to the top of M_U and one atom to the end of M_D, and join the two new atoms. The old lower endpoint changes from two bonds and two free ends to three bonds and one free end, so its degree is unchanged. The new upper endpoint is removed last; therefore the transpose phase and all its k cleanups are unchanged. The last lower atom then has degree three, and its pair with the remaining upper atom is one operation (b). For $n = 3k + 2$,

$$(A, B, C, D, U) = (k + 1, 1, 0, k, 2k), \qquad \#\{33\} = k + 1.$$

Trimmed family. For $k \geqslant 2$, take $M_U = (u_0, \dots, u_{3k-1})$ and $M_D = (d_0, \dots, d_{3k-2})$ as paths. Use the transpose cross-bonds except that the bond which would end at d_{3k-1} is redirected to d_0; the extra u_{3k} is omitted. The first k upper cuts and the next $k - 1$ block cleanups are the same as above. The cascade stops at the tail pair d_{3k-3}, d_{3k-2}: one further operation (a) lowers the latter to degree three, after which the pair is a $\{33\}$ cleanup. This final cleanup breaks one live cross-bond, while each preceding block breaks two. The remaining $2k - 1$ upper atoms are then $\mathrm{CUT_U}$ atoms, and

$$(A, B, C, D, U) = (k + 1, 0, 1, k - 1, 2k - 1), \qquad \#\{33\} = k.$$

Together with Theorem 2.1, these families prove

$$a_n = \left\lceil \frac{n - 1}{3} \right\rceil, \qquad n \geqslant 4. \tag{3.1}$$

4 Intrinsic consequence on Toy I

The benchmark theorem supplies one legal witness in the intrinsic maximum; it does not assert that every legal cutting sequence has the same count.

Proposition 5 (Scoped intrinsic lower bound)**.** *Fix $d > 0$. For every finite Toy I molecule with both layers nonempty,*

$$v_d(M) \geqslant \left\lceil \frac{|M_U| - 1}{3} \right\rceil - 10d = \left\lceil \frac{\rho(M)}{3} \right\rceil - 10d. \tag{4.1}$$

Proof. Let $n = |M_U|$ and $m = |M_D|$. The two layer trees have $n - 1$ and $m - 1$ internal edges, and Toy I has exactly n cross-bonds. The full graph is connected. Here $\rho(M)$ denotes its circuit rank, $\rho(M) = |E| - |V| + 1$, so

$$\rho(M) = |E| - |V| + 1 = ((n - 1) + (m - 1) + n) - (n + m) + 1 = n - 1.$$

The source algorithm supplies a legal complete run with $\#\{4\} = 1$, and Theorem 2.1 gives $\#\{33\} \geqslant \lceil (n-1)/3 \rceil$ for it. Since v_d is the maximum score over all legal complete cuts, it is at least the score of this witness. □

For $n \geqslant 4$, define the finite-size constant by an infimum (the Toy I class has no a priori bound on $|M_D|$):

$$c_d^{\mathrm{ToyI}}(n) = \inf_{\substack{M \text{ finite Toy I} \\ |M_U|=n}} \frac{v_d(M)}{\rho(M)}.$$

Then

$$c_d^{\mathrm{ToyI}}(n) \geqslant \frac{\lceil (n-1)/3 \rceil - 10d}{n-1}, \qquad \liminf_{n\to\infty} c_d^{\mathrm{ToyI}}(n) \geqslant \frac{1}{3}. \tag{4.2}$$

This is an asymptotic statement within Toy I, not the global infimum c_d^*.

5 Audit and reproducibility

The proof uses no finite enumeration. A bounded full-labelled replay checks 11,590 molecules and 25,243 complete tie-break records with total size at most six, including double-parent configurations in both layers. It reports no failed records and no violations of the local flow contract. This replay is regression evidence only.

The public code and reproducibility repository is available at:

https://github.com/xingzhicn/molecule-cut

It reproduces the exact subset-DP values and the family profiles. The reproducibility commands are

```
uv run --frozen pytest
uv run --frozen ruff check .
```

On August 5, 2026 these checks reported 652 passing tests, a clean Ruff check, and all source hashes valid.

6 What is and is not concluded

We prove the exact Toy I/Definition-11.4 benchmark constant 1/3, its explicit sharpness families, and the scoped intrinsic consequences (4.1)–(4.2). We do not prove a global positive lower bound for c_d^*, a theorem for W, a result for Toy I plus/II/III or the full molecule, a score bound for every arbitrary legal sequence, or any complexity-hardness classification. No claim is made about global no-localization, no-beam, or impossibility of the broader DHM/Boltzmann program.

References

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[3] Y. Bruned and V. Clarisse, *Kruskal-style algorithm for cubic Schrödinger equation molecule reduction*, arXiv:2603.23298, 2026.